📋 Product Requirement Document (PRD) & Technical Specification

Project: SLICE (`slice`) — Multi-Tenant System Container & Private Cloud VPS Engine

> **Document ID:** PRD-SLICE-2026-V1
> **Classification:** Confidential / Internal Engineering Blueprint
> **Project Code:** *`slice`* (formerly hvps)
> **Target Node:** *`playfam1`* (Tailscale: *`100.108.185.46`* · LAN Wi-Fi: *`192.168.20.193`*)
> **Hardware:** HP Pavilion Gaming Laptop 15-dk1xxx (Intel Core i7-10750H · 16GB RAM · 1TB SATA · GTX 1650)
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> **Release Date:** November 2026 · Version 1.0.0

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1. 🎯 Executive Summary & Vision

1.1 Context & Background

The expansion of our computing infrastructure across Telkom OPMC and the ATNX fleet has reached three dedicated nodes: 1. **`fm.atnx` (ASUS TUF F15):** Central operational command center, hosting core databases, Coolify, 9router, and the primary ATNX autonomous agent. 2. **`playfam` (Desktop PC i7-4790):** Dedicated host for **Hermes Box (H-Box)**, serving multi-tenant sandbox environments tailored for beginner office peers (strict 1.5 GB limits, containerized Hermes AI, and integrated 24/7 Telegram bots). 3. **`playfam1` (HP Pavilion Gaming 15):** Designated as the **Advanced Developer Fleet Node**.

Advanced software engineers in our circle require full freedom that conventional application containers (plain Docker) cannot offer without severe trade-offs. They require: * True operating system environments with native **`systemd`** as PID 1 (`systemctl` , `journalctl` , daemon supervisors). * Full rootless and nested containerization capabilities (**Docker-in-VPS / Docker Compose** within their own sandbox). * Unrestricted package installation (`apt` , `snap` , custom compilers for Rust, Go, Python, Node.js). * Dedicated development port ranges to expose internal web servers, microservices, and databases for testing. * Optional hardware acceleration via the onboard **NVIDIA GeForce GTX 1650 (4GB GDDR6)** for local AI inference, CUDA tasks, and media processing.

1.2 The Solution: `slice`

`slice` is an ultra-lightweight, high-performance Private Cloud VPS manager built on top of **Incus** (the independent, community-driven continuation of Canonical's LXD by the Linux Containers team).

Instead of heavy hardware emulation (KVM / Proxmox) which locks RAM statically and wastes compute cycles, `slice` provisions **System Containers** running Ubuntu 24.04 LTS. Each instance behaves 100% like a dedicated cloud VPS from DigitalOcean, Linode, or Hetzner, while operating with near-zero virtualization overhead and shared kernel efficiency.

1.3 Why the Name "Slice"?

Historically, before the term "VPS" became generic, pioneer hosting platforms coined the term "a slice of a server" (most notably Slicehost). The name `slice` captures this philosophy perfectly: * **Monosyllabic & Ergonomic:** Single-syllable, punchy, memorable, and lightning-fast to type in terminal sessions (`slice list` , `slice create`). * **Universal & Non-Biased:** Free of vendor or platform constraints (unlike hvps or hbox). A slice is simply a clean, independent slice of modern Linux computing power for developers to use as they see fit.

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2. 🖥️ Physical Hardware Baseline (`playfam1`)

Comprehensive hardware discovery performed on node `playfam1` confirms the following physical topology:

Hardware Subsystem	Detected Specification	Engineering Assessment
Chassis / Model	HP Pavilion Gaming Laptop 15-dk1xxx	Dual-fan rear exhaust thermal chassis
Processor (CPU)	Intel Core i7-10750H (Comet Lake-H)	6 Cores / 12 Threads @ 2.60 GHz (Max Turbo 5.0 GHz) · 12MB Cache · Hardware VT-x Enabled
System Memory (RAM)	16 GB DDR4-2933 MHz (15 GiB usable)	4.0 GB Swap space active on drive. Base idle consumption: ~820 MB
Primary Storage (SATA)	1.0 TB SATA HDD/SSD (`/dev/sda2` : 930.5 GB)	Mounted at `/` . 860 GB available unallocated space. Selected as the **exclusive** `slice` storage pool
Secondary Storage (NVMe)	256 GB M.2 NVMe SSD (`/dev/nvme0n1`)	Contains pre-installed factory OEM Windows (NTFS, EFI, Recovery). **STRICTLY EXCLUDED / UNTOUCHED**
Discrete GPU	NVIDIA GeForce GTX 1650 Mobile / Max-Q	Chipset TU117M (4GB GDDR6 VRAM) · 896 CUDA Cores · NVENC/NVDEC hardware engines
Operating System	Ubuntu Server 24.04.5 LTS (Noble Numbat)	Linux Kernel 6.8.0-149-generic (x86_64)
Network Interfaces	Tailscale P2P (`100.108.185.46`) + Wi-Fi	Sub-30ms latency over encrypted Tailscale mesh

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3. 🏗️ Core Technology: Why Incus System Containers?

Traditional virtual machines (KVM) emulate physical hardware (virtual BIOS, virtual PCI buses, virtual NICs), incurring an unavoidable 10%–15% memory and CPU tax. Traditional Docker containers, on the other hand, isolate single processes and lack `systemd` , making it frustrating for developers who need to run background services, databases, and cron daemons together.

Incus bridges this gap:

```
┌──────────────────────────────────────────────────────────┐
│                  ARCHITECTURE COMPARISON MATRIX          │
├──────────────────────────┬──────────────────────────┬────┤
│ Docker Application Ctr   │ Incus System Container   │ Hardware KVM / VM │
├──────────────────────────┼──────────────────────────┼──────────────────┤
│ Single Process (PID 1)   │ Full Init System (systemd│ Full OS & Kernel  │
│ No background daemons    │ Native daemons & services│ Full daemons      │
```

```
| High risk for nested Dkr| Native secure Docker DinI| Secure nesting    |
| Shared root risk        | Unprivileged User Namespc| Hard hypervisor   |
| RAM overhead: ~10 MB    | RAM overhead: ~25-35 MB  | RAM overhead: ~1GB|
| 16GB limit: Many apps   | 16GB limit: 3-5 Full VPS | 16GB limit: 2 VM  |
└─────────────────────────┴──────────────────────────┴───────────────────┘
```

3.1 Security via Unprivileged User Namespaces

By default, all Incus containers run as **unprivileged containers**. The container's root user (`UID 0`) is mapped to an unprivileged sub-UID on the host (e.g., `UID 1000000`). Even if a malicious actor or errant script achieves root privilege inside a slice, their real UID on the host laptop remains unprivileged, eliminating container breakouts to the host filesystem.

3.2 Native Docker Nesting (`security.nesting=true`)

Developers can install Docker Engine, Docker Compose, or lightweight Kubernetes (`k3s`) directly inside their slice. Incus configures the necessary cgroup controllers and namespace mounts safely without requiring dangerous privileged flags.

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4. 🗄️ Storage Architecture & Disk Policy

4.1 Strict Preservation of OEM Windows NVMe

The secondary 256 GB NVMe SSD (`/dev/nvme0n1`) holds: * `/dev/nvme0n1p1` : EFI System Partition (`vfat` , UUID `2050-A441`) * `/dev/nvme0n1p2` : Microsoft Reserved Partition * `/dev/nvme0n1p3` : Windows Basic Data Partition (`ntfs` , 237.9 GB) * `/dev/nvme0n1p4` : Windows Recovery Environment (`ntfs`)

Policy Directive: Under no circumstances shall `/dev/nvme0n1` be formatted, mounted, repartitioned, or added to Incus storage pools. This guarantees that the laptop can revert to Windows or dual-boot at any time without data loss.

4.2 Primary SATA Pool Configuration

All system containers, root filesystems, images, and user volumes reside on `/dev/sda2` : * **Pool Name:** `slice-pool` * **Driver:** `dir` or loop-backed `btrfs` / `zfs` on `/var/lib/incus/storage-pools/slice-pool` . * **Available Capacity:** ~860 GB free space. * **Storage Allocation per Slice:** Hard capped at **200 GB per user**, enforced via quota mechanisms. * **Host Cushion:** ~260 GB retained exclusively by the host OS for caching, temporary compile artifacts, and system updates.

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5. 📊 Capacity Sizing & Resource Model (3-Instance Model)

To guarantee high responsiveness, thermal safety for the laptop, and genuine "pro-tier" performance for developers, the resources of `playfam1` are partitioned according to a strict **1 Host + 3 Slices** formula:

```
┌──────────────────────────────────────────────────────┐
│              PHYSICAL HARDWARE: HP Pavilion 15 (16GB RAM)              │
└──────────────────────────────────────────────────────┘
                                 │
                 ┌───────────────┼──────────────┐
                 ▼               ▼              ▼
┌──────────┐  ┌──────────┐  ┌──────────┐
│  HOST OS │  │  SLICE 1 │  │  SLICE 2 │ ... (SLICE 3)
│ 3.0 GB RAM│  │4.0 GB RAM │  │4.0 GB RAM │
│ 2 Threads │  │ 4 vCPU   │  │ 4 vCPU   │
└──────────┘  └──────────┘  └──────────┘
```

```
| ~260 GB  |                | 200 GB   |                | 200 GB   |
└──────────┘                └──────────┘                └──────────┘
```

5.1 Resource Allocation Matrix

Instance / Entity	Guaranteed RAM	CPU Allocation	Disk Quota	Process Cap	Roles & Capabilities
🖥️ **Host OS (`playfam1`)**	**3.0 GB**	**2 Core Reserve** (Kernel Prio)	**~260 GB**	**Kernel stability**, I/O disk buffer cache, Tailscale daemon, Incus management.	
🍰 **Slice 1 (Dev A)**	**4.0 GB**	**4 vCPU** (Dynamic Burst)	**200 GB**	**500 PIDs**	Full VPS sandbox. Docker, Node.js, Go, Rust, PostgreSQL, Redis.
🍰 **Slice 2 (Dev B)**	**4.0 GB**	**4 vCPU** (Dynamic Burst)	**200 GB**	**500 PIDs**	Full VPS sandbox. Docker, Node.js, Go, Rust, PostgreSQL, Redis.
🍰 **Slice 3 (Dev C)**	**4.0 GB**	**4 vCPU** (Dynamic Burst)	**200 GB**	**500 PIDs**	Full VPS sandbox. Docker, Node.js, Go, Rust, PostgreSQL, Redis.
TOTAL	**15.0 GB**	**12 vCPU Total**	**~860 GB**	—	**Zero waste, mathematically balanced, 100% capacity utilization.**

5.2 Technical Justification

1. **3.0 GB Host Reservation:** When three developers concurrently execute intensive tasks (such as `npm install`, `cargo build`, or compiling Docker images), disk I/O demands skyrocket. The Linux kernel relies on unallocated physical memory as page cache to smooth out disk write spikes. Reserving 3.0 GB prevents laptop freezing, audio/system jitter, and kernel OOM panics.
2. **Dynamic CPU Burst (4 vCPU Limit):** Developer workloads are inherently spiky. Instead of statically pinning 3 threads per instance, Incus assigns `limits.cpu: 4` with balanced CFS shares. A developer compiling code can consume up to 4 full threads instantly, releasing them back to the pool immediately upon completion.
3. **Generous 200 GB Storage:** Commercial cloud providers charge substantial premiums for 100GB+ SSD storage. Offering 200 GB per slice provides ample headroom for Docker images, local databases, and large datasets without friction.

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6. 🌐 Network Architecture & Ingress

6.1 Internal Bridge (`slicebr0`)

- Incus provisions an isolated virtual bridge: `slicebr0`.
- IPv4 Subnet: `10.150.0.1/24` with internal dnsmasq handling DHCP and hostname resolution.
- Slices receive deterministic private IPs: `10.150.0.11`, `10.150.0.12`, `10.150.0.13`.

6.2 Port Forwarding Architecture

Because developers access `playfam1` across Tailscale or office Wi-Fi, port forwarding is managed via Incus native proxy devices (`type: proxy`), bypassing complex manual iptables:

A. SSH Access Ports

- **Slice 1:** `100.108.185.46:2211` ⟶ Slice 1 Port `22`
- **Slice 2:** `100.108.185.46:2212` ⟶ Slice 2 Port `22`
- **Slice 3:** `100.108.185.46:2213` ⟶ Slice 3 Port `22`

B. Development Port Blocks (Web Apps & API Testing)

Each slice receives a dedicated block of 5 forwarded ports to test web applications without port collisions: * **Slice 1:** Host Ports `3101 - 3105` ⟶ Container Ports `3101 - 3105` * **Slice 2:** Host Ports `3201 - 3205` ⟶ Container Ports `3201 - 3205` * **Slice 3:** Host Ports `3301 - 3305` ⟶ Container Ports `3301 - 3305`

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7. 🎮 GPU Passthrough (NVIDIA GTX 1650 4GB VRAM)

Node `playfam1` houses an onboard **NVIDIA GeForce GTX 1650 Mobile (TU117M)**. Incus supports sharing host GPU devices with unprivileged containers without requiring full VFIO PCI stubbing.

7.1 Passthrough Implementation

Host NVIDIA proprietary drivers are installed on `playfam1` . When a developer requires GPU power (for PyTorch, Whisper, or 3D rendering), the administrator attaches the GPU device dynamically:

```
slice gpu attach <slice_name>
```

Which executes the underlying Incus call:

```
incus config device add <slice_name> gpu0 gpu id=01:00.0
```

7.2 Developer Experience

Inside the slice, the developer can run `nvidia-smi` natively, access CUDA compute capabilities, and utilize NVENC hardware acceleration with zero emulation overhead.

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8. 🔋 24/7 Laptop Hygiene & Reliability Protocols

Operating a gaming laptop as a 24/7 headless server requires proactive thermal and ACPI management to prevent unwanted sleep states:

1. **Lid Close Inhibition (`/etc/systemd/logind.conf`):** `ini [Login] HandleLidSwitch=ignore HandleLidSwitchExternalPower=ignore HandleLidSwitchDocked=ignore LidSwitchIgnoreInhibited=no`
2. **Masking System Suspend & Sleep Targets:** `bash sudo systemctl mask sleep.target suspend.target hibernate.target hybrid-sleep.target`
3. **Physical Ventilation:** The laptop must be mounted on a ventilated riser or stand (rear lifted by at least 2.5 cm) to facilitate unrestricted intake airflow for the dual-fan cooling assembly.

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9. 📁 Project Structure & Filesystem Layout

All code, configurations, templates, and state tracking for the `slice` platform will be consolidated under **/srv/slice/**:

```
/srv/slice/
├── bin/
│   └── slice                    # Primary Bash CLI orchestrator (symlinked to /usr/local/bin/slice)
├── profiles/
│   ├── slice-standard.yaml      # Standard Incus profile (4GB RAM, 4 vCPU, 500 PIDs, nesting=true)
│   └── slice-gpu.yaml           # GPU passthrough profile overlay
├── templates/
│   ├── user-data.yaml           # Cloud-init template (provisions user, passwordless sudo, sshd config)
│   └── motd-banner.sh           # Elegant terminal welcome banner shown on SSH login
├── state/
│   └── slice-state.json         # Local database tracking assigned ports, IPs, quotas, and users
├── scripts/
│   ├── 01-host-prep.sh          # Idempotent script for lid sleep masking and kernel adjustments
│   ├── 02-init-storage.sh       # Formats/initializes SATA storage pool and incusbr0
│   └── 03-install-incus.sh      # Automates Incus package installation and group permissions
├── docs/
│   └── PRD_SLICE_VPS.pdf        # Compiled technical PRD and architecture blueprint
└── README.md                    # Operational runbook for administrators
```

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10. 🛠️ CLI Manager Specification: `slice`

The `/usr/local/bin/slice` command provides an intuitive, high-level control plane for administrator operations.

10.1 Command Vocabulary

Command Syntax	Description	Example Usage
`slice create <name> [port] [pass]`	Provisions a new Ubuntu 24.04 VPS slice with 4GB RAM, 4 vCPU, and 200GB storage.	`slice create danz`
`slice list` (or `slice ls`)	Displays a formatted table of all active slices, IP addresses, ports, and resource usage.	`slice list`
`slice stats` (or `slice top`)	Live telemetry dashboard showing CPU, RAM, and disk utilization per slice.	`slice stats`
`slice invite <name>`	Generates a clean, ready-to-copy WhatsApp onboarding card with credentials and ports.	`slice invite danz`
`slice gpu attach <name>`	Grants the slice direct access to the NVIDIA GTX 1650 GPU.	`slice gpu attach danz`
`slice gpu detach <name>`	Reclaims GPU access back to the host.	`slice gpu detach danz`
`slice port add <name> <port>`	Forwards an additional custom port from host to slice.	`slice port add danz 8080`
`slice port list <name>`	Lists all active port forwardings for a specific slice.	`slice port list danz`
`slice enter <name>`	Opens an instant interactive root shell inside the container (emergency rescue).	`slice enter danz`

Command Syntax	Description	Example Usage
`slice start / stop / restart`	Controls the lifecycle of a slice container.	`slice restart danz`
`slice delete <name> [--purge]`	Destroys the slice instance (optionally wiping persistent data).	`slice delete danz --purge`

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11. 📩 Onboarding & Developer Experience

Upon provisioning via `slice create <name>`, the administrator executes `slice invite <name>`, producing a formal, professional invitation card:

```
======================================================================
  📤 PANDUAN AKSES LINGKUNGAN PRIVATE CLOUD VPS: 'SLICE'
======================================================================
Yth. Rekan Pengembang / danz,

Lingkungan Private VPS mandiri Anda pada node 'playfam1' telah aktif.

Spesifikasi Sumber Daya (Dedicated Slice):
• Sistem Operasi         : Ubuntu 24.04 LTS (Native systemd)
• Alokasi Memori (RAM)   : 4.0 GB Dedicated (Hardcap)
• Alokasi Prosesor (CPU) : 4 vCPU (Dynamic Burst)
• Alokasi Penyimpanan    : 200 GB Storage (High-Speed SATA)
• Akses Administratif    : Full Sudo / Root Access di dalam VPS
• Dukungan Docker        : Native Docker & Docker Compose Enabled

Informasi Akses SSH:
• Host Jaringan (Tailscale) : 100.108.185.46
• Host Jaringan (Wi-Fi)     : 192.168.20.193
• Port SSH Layanan          : 2211
• Nama Pengguna (Username)  : danz
• Kata Sandi Sementara      : [REDACTED]

Jatah Port Development (Testing Web / API / DB):
• Port Forward Aktif : 3101 s/d 3105 (Host ⟶ VPS Anda)

Perintah Menghubungkan (Terminal Mac / Linux / PowerShell):
ssh danz@100.108.185.46 -p 2211

Catatan Keamanan:
Disarankan untuk segera mendaftarkan SSH Public Key Anda pada file
~/.ssh/authorized_keys dan memperbarui kata sandi awal Anda.
======================================================================
```

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12. 🚀 Phased Implementation Roadmap

- **Phase 1: Host Preparation & Hygiene (ETA: Day 1)**
- Implement lid switch ignore and mask sleep targets in systemd.
- Verify network parameters and Tailscale routing.
- **Phase 2: Incus Engine & Storage Initialization (ETA: Day 1)**
- Install Incus from official Linux Containers repositories.
- Initialize `slice-pool` on `/dev/sda2` (200GB quota capabilities).
- Provision virtual bridge `slicebr0`.
- **Phase 3: Directory Blueprint & CLI Engineering (ETA: Day 2)**
- Construct `/srv/slice/` workspace.

- Implement `/usr/local/bin/slice` orchestrator script with full vocabulary.
- Configure cloud-init automation templates and unprivileged profiles.
- **Phase 4: Pilot Testing & Verification (ETA: Day 2)**
- Spin up pilot slice `slice-pilot` on port 2211.
- Test SSH login, `systemctl` service management, and nested Docker installation.
- Validate 4GB RAM ceiling and dynamic CPU limits.
- **Phase 5: Production Handover (ETA: Day 3)**
- Provision slices for designated developer peers.
- Dispatch onboarding cards and conduct orientation.

SLICE: High-Performance Private Cloud VPS Infrastructure
Designed & Maintained by ATNX 🖤⚡ · Node playfam1 · Telkom OPMC